Lighting a Line-Scan Camera: Illumination Design and Camera Calibration

Juan Carlos Chavarría Morales
M.Sc. Optimization, Hochschule Bielefeld (HSBI)
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Abstract

A line-scan camera images one line of a moving surface; for a clean image, the light along that line must be as uniform as possible. We model the illumination of a laboratory set-up, two rectangular light panels at the ends of a scan line about half a metre long, and optimise their placement. Three results. First, the objective used in the project's first scripts, the absolute standard deviation of the irradiance, is minimised by dimming the lights: a relative measure (min/mean or standard deviation/mean) is needed. Second, at the same efficiency as the lab geometry, a different placement of the panels raises the uniformity $U_0 = \min/\text{mean}$ along the line from 0.80 to 0.95, and the Pareto front shows how much uniformity each gain in efficiency costs; in our reading of the geometry, the lab panels also slightly enter the camera's field of view. Third, from seven line-scan images of a checkerboard target we calibrate the camera geometrically: 0.120 mm per pixel on the target, residuals of 0.06 px once the printing errors of the target (±0.14 mm) are estimated jointly. The same images cannot validate the illumination model: the light areas are saturated. We give the measurement protocol that would.

1 The set-up

The scan line is the x-axis of the inspection plane $z = 0$. A line-scan camera with 4096 pixels looks down from $z_c = 606.5\,\text{mm}$; its field of view on the target is 491 mm (Section 4). Two light panels of 210 mm × 510 mm are placed symmetrically at the ends of the line: centres at $x = \pm W/2$, $W = 480\,\text{mm}$, height $z_0 = 209\,\text{mm}$, tilted by $\theta = 15°$ (Figure 2, left). The project asked how to place such lights for a uniform line, and how to calibrate the camera.

2 Models and the measure of uniformity

Point sources. The first model of the project replaced each light by an isotropic point source of power P at $(\pm d, z_s)$; with the inverse-square law and the cosine of incidence,

$$E(x) = \frac{P z_s}{((x-d)^2 + z_s^2)^{3/2}} + \frac{P z_s}{((x+d)^2 + z_s^2)^{3/2}}. \tag{1}$$

Panels. For the lab, each panel is a generalised Lambertian emitter (Moreno and Sun, 2008): a surface element $\mathrm{d}A$ at $\mathbf{q}$ with normal $\mathbf{n}$ sends to the point $\mathbf{p}$ on the line

$$\mathrm{d}E = L_0 \frac{\cos^m \vartheta_s \, \cos \vartheta_r}{\|\mathbf{p} - \mathbf{q}\|^2} \, \mathrm{d}A, \tag{2}$$

with ϑ_s the emission angle at the panel, ϑ_r the incidence angle on the plane, and $m = 1$ for a diffuse (Lambertian) panel, $m > 1$ for a more directed one. The irradiance is the integral over

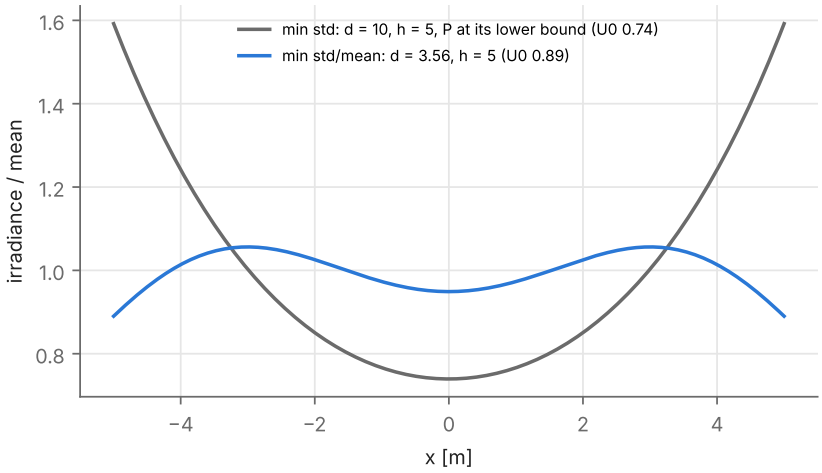


Figure 1: Two point sources on a 10 m line: the optimum of the absolute standard deviation (the sources dimmed and moved to the bounds) and of the relative one.

both panels, computed by Gauss–Legendre quadrature (8×12 nodes per panel are converged to 10^{-6}; the 3×3 nodes of the original notebook give errors of up to 6 %).

Uniformity. We use the overall uniformity of lighting standards, $U_0 = E_{\min}/\bar{E}$, and the coefficient of variation $\mathrm{CV} = \sigma_E/\bar{E}$. Both are relative. The first scripts of the project minimised the absolute standard deviation σ_E over the powers, the height and the distance of two point sources. Since σ_E is proportional to the power, the optimiser lowers the power to its bound and moves the lights as far away as allowed: the "optimum" sits on the bounds and has $U_0 = 0.74$. Minimising CV instead gives $U_0 = 0.89$ for the same bounds (Figure 1). The same issue reappears in any design problem in which the objective scales with the light: uniformity must be measured relative to the mean, and the light output treated as a separate goal.

3 Designing the illumination

Over the field of view, the lab set-up has $U_0 = 0.80$: the line is 20 % darker in the middle than on average (Figure 2, right). Tilting the panels further towards the centre helps: at 25°, $U_0 = 0.83$ with the same light on the line. In our reading of the geometry, the inner edges of the lab panels reach about 1 cm into the cone in which the camera sees the line; from 25° on they are out of it. This should be checked on the real set-up.

Optimisation. What can be chosen is the geometry: distance W, height z_0 and tilt θ. Two goals conflict: uniformity, and efficiency, i.e. how much of the emitted flux reaches the line (mean irradiance per unit emitted flux, relative to the lab set-up). We maximise U_0 for a range of minimum efficiencies, with the constraint that the panels stay out of the camera's view, by differential evolution with a local polish. Table 1 and Figure 3 give the Pareto front. At the efficiency of the lab set-up, U_0 rises from 0.80 to 0.95 (panels closer together, higher and tilted by 30°); a uniformity of 0.97 to 0.99 is possible at 80 to 45 % of the efficiency; and 20 % more efficiency costs uniformity down to 0.85. More directed emitters ($m = 3$) give a similar front, slightly lower at high efficiency.

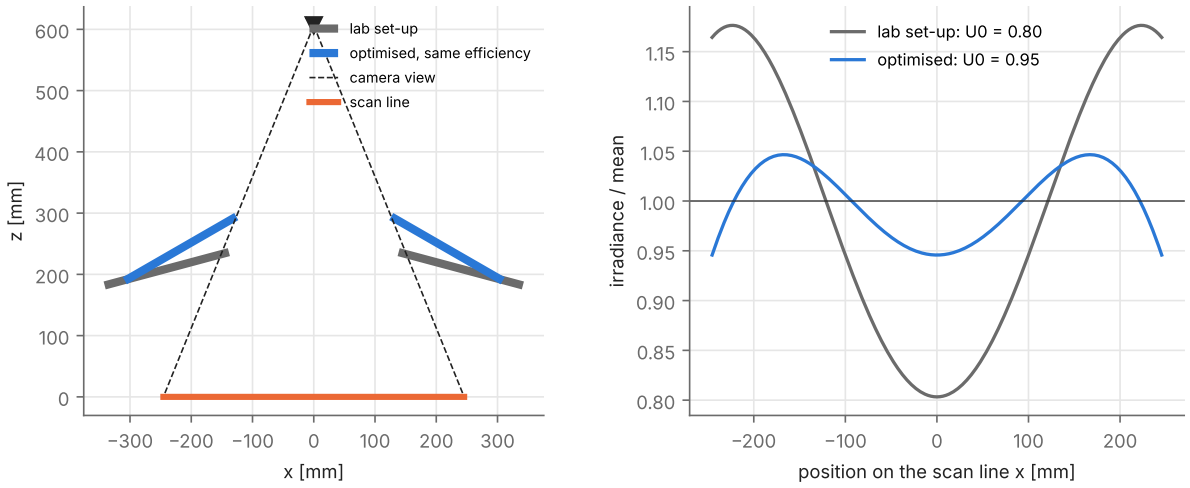


Figure 2: Left: the lab set-up and the optimised placement of the panels at the same efficiency (section through the scan line; the panels extend 510 mm across it), with the camera's view of the line. Right: irradiance along the line, normalised by its mean.

Table 1: Lab set-up and optimised designs (Lambertian panels, field of view 491 mm). Efficiency relative to the lab set-up.

design	W [mm]	z_0 [mm]	θ [°]	U_0	CV	efficiency	blocks view
lab set-up	480	209	15	0.80	0.137	1.00	yes
lab, tilt 25°	480	209	25	0.83	0.115	1.02	no
optimum, efficiency ≥ 0.4	636	450	33	0.989	0.007	0.45	no
optimum, efficiency ≥ 0.6	512	386	13	0.982	0.011	0.60	no
optimum, efficiency ≥ 0.8	455	308	15	0.969	0.020	0.80	no
optimum, efficiency ≥ 1.0	435	242	30	0.946	0.035	1.00	no
optimum, efficiency ≥ 1.2	415	157	55	0.851	0.103	1.20	no

4 Calibrating the line-scan camera

A line-scan image repeats one sensor line over time; seven images show a row of eight edges of a checkerboard target (square size 26.54 mm) at different positions (Figure 4). Averaging the 50 lines of an image and locating the mid-level crossings with sub-pixel interpolation gives eight edge positions per image. Two images were taken at the same position: their edges agree to 0.024 px (RMS), which is the repeatability of the measurement.

A line camera maps a position X on a (possibly tilted) target to the pixel $u = (a+bX)/(1+cX)$, a 1-D projective map (the general line-scan model is discussed by Steger and Ulrich, 2021). Per image, a linear map leaves residuals of 1.4 px (RMS) and the projective map 0.6 px (Table 2 and Figure 5), 25 times the repeatability. The residual pattern, however, is the same in every image, although the target lies at different places on the sensor: it belongs to the target, not to the camera. Estimating one deviation per edge of the target, common to all images, together with the per-image maps, reduces the residuals to 0.06 px for six images, about 7 µm on the target. The printed squares deviate from their nominal positions by up to 0.14 mm (Figure 5, right).

Results. The scale on the target is 8.34 px/mm, i.e. 0.120 mm per pixel and a field of view of 491 mm; across the target it changes by 0.2 to 2.7 %, a sign of a slightly tilted target. Three limits: (i) the edges cover only pixels 1000 to 2950, so lens distortion near the ends of the sensor cannot be identified; (ii) the image taken with a different exposure (00000037) does not fit the others (0.38 px): in saturated images the position of an edge depends on the exposure; (iii) the

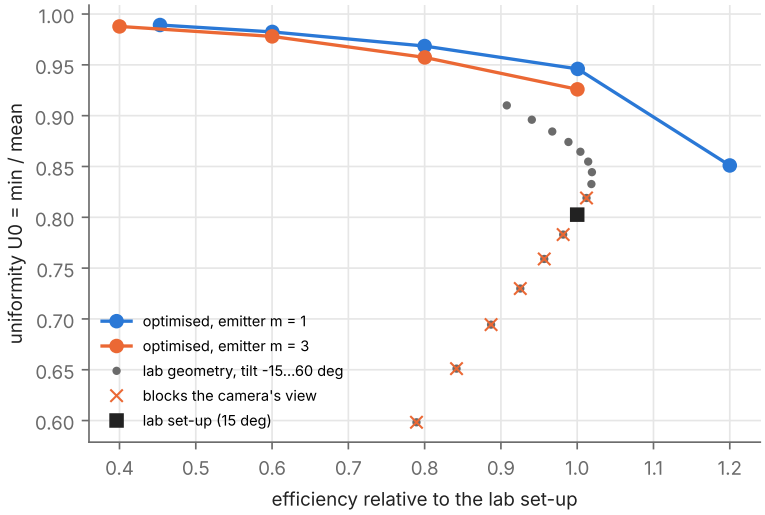


Figure 3: Uniformity against efficiency: Pareto fronts of the optimised geometry, and the lab geometry with different tilts (crosses: the panels enter the camera's view). For $m = 3$ the efficiency is relative to the lab geometry with that emitter.

Table 2: Calibration per image: share of saturated pixels, range of the edges on the sensor, scale, and RMS residuals of the three models (the joint fit with target deviations excludes image 00000037, taken with a different exposure).

image	saturated [%]	edges [px]	scale [px/mm]	RMS residual [px]		
				linear	projective	+ target
00000028	34	1206–2755	8.343	1.22	0.57	0.048
00000029	34	1199–2748	8.345	1.28	0.60	0.035
00000030	34	1253–2802	8.348	1.11	0.60	0.035
00000032	35	1004–2550	8.330	2.00	0.61	0.041
00000033	34	1004–2550	8.330	1.99	0.61	0.042
00000034	42	1402–2952	8.349	0.71	0.69	0.118
00000037	14	1182–2732	8.346	1.39	0.77	–

intrinsic matrix of the original notebook came from an area-camera calibration routine applied to these images. Its vertical entries have no meaning for a line-scan camera (the vertical axis is time), and its focal length (4804 px) together with the measured scale implies a working distance of 576 mm, 5 % less than the camera height used in the model.

5 What has not been validated

The images were meant to validate the illumination model as well, by fitting the emission parameters of the panels to the measured profile. They cannot: in six of them 34 to 42 % of the pixels are at the maximum grey level (14 % in the seventh), the light areas carry no information about the irradiance, and any uniformity computed from them is close to 1 by construction. The original notebook also maximised the uniformity over the emission parameters instead of fitting them: emission is a property of the panels, to be measured; only the geometry is to be optimised. A validation would need:

1. a diffuse white target (e.g. PTFE) without pattern, imaged with exposure and gain set so that the brightest pixel stays below about 80 % of the range, plus a dark frame;

2. the camera's flat-field response (vignetting), measured once under uniform light, divided

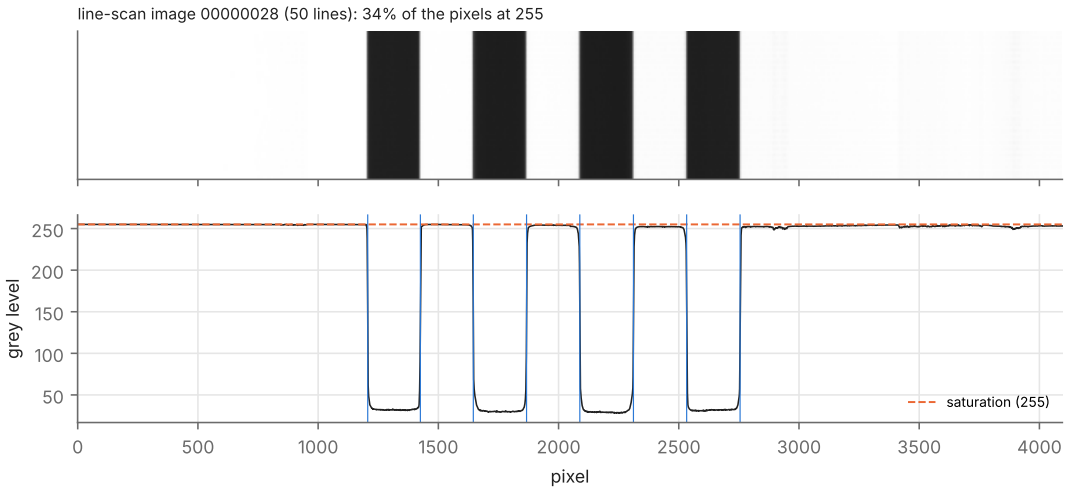


Figure 4: A line-scan image of the target and its mean profile. The light squares and the background are saturated (grey level 255); blue: detected edges.

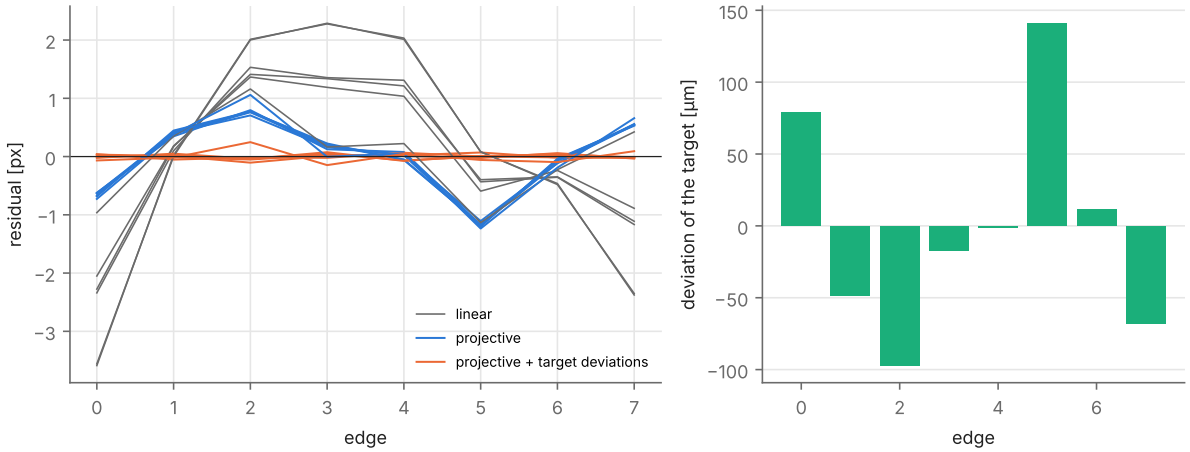


Figure 5: Left: residuals of the linear, projective and joint models for the six images with the same exposure. Right: estimated deviations of the target's edges from the nominal grid (the part not absorbed by a tilt or scale of the projective map).

out;

3. a least-squares fit of the emission exponent m and the scale L_0 to the profile, and only then the design optimisation of Section 3, followed by a measurement at the optimised geometry.

6 Main changes

Table 3: Main changes with respect to the original scripts and notebooks.

	original	this version
Uniformity	absolute standard deviation (minimised by dimming); later min/mean on saturated data	U_0 and CV on the model; efficiency as a separate goal
Panel model	beam profile $1/(1+(r/a)^2)^b$ without cosines, 3×3 quadrature	generalised Lambertian emitter with emission and incidence cosines, converged quadrature
Optimisation	of the emission parameters (properties of the hardware)	of the geometry, with a view constraint; Pareto front uniformity vs efficiency
Calibration	area-camera routine on line-scan images	1-D projective model per image, joint estimation of the target's printing errors
Validation	claimed from saturated images	not possible with these images; protocol given
Code	notebooks	package, scripts for every table and figure, tests on synthetic data

7 Conclusions

1. Uniformity has to be measured relative to the mean light level; minimising the absolute spread optimises the lights away.
2. For the lab panels, a different placement (closer together, higher, more tilted) raises the uniformity along the line from 0.80 to 0.95 without losing light; the uniformity–efficiency trade-off is quantified by the Pareto front. The lab panels seem to touch the camera's view and should be checked.
3. The line-scan camera is calibrated to 0.120 mm per pixel with residuals of 0.06 px, once the printing errors of the target are estimated; a target spanning the whole sensor would also identify lens distortion.
4. The illumination model is not yet validated: the existing images are saturated. A measurement with a plain white target at lower exposure is the next step, and the deciding one for whether the idea is worth pursuing.

Code and data

Code, the seven calibration images and the scripts for every table and figure: Chavarría Morales (2026).

References

Juan Carlos Chavarría Morales. linescan-lighting: illumination design and calibration for a line-scan camera. https://github.com/JuanCaChavaMor/linescan-lighting, 2026.

Ivan Moreno and Ching-Cherng Sun. Modeling the radiation pattern of LEDs. *Optics Express*, 16(3):1808–1819, 2008.

Carsten Steger and Markus Ulrich. A camera model for line-scan cameras with telecentric lenses. *International Journal of Computer Vision*, 129(1):80–99, 2021. doi: 10.1007/s11263-020-01358-3.